

THE LIVING MAP

Spatial Memory for Emergency Robots

TSYP14 Technical Challenge • Phase 1 Technical Proposal



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Spatial Memory for Emergency Robots

TSYP14 Technical Challenge | IEEE RAS + IEEE AESS Tunisia Section Chapters

Phase 1 Proposal A two-robot system that gives a GPS-denied, disconnected environment a persistent memory using autonomous radio beacons.

1. Challenge and motivation

In disaster response, robots may operate inside collapsed, fire-damaged, underground or contaminated environments where GPS and communication are unavailable. The challenge is not only to explore such spaces, but to preserve critical knowledge so that a later robot can continue the mission without starting from zero.

Our Phase 1 working scenario is a Fire / Hazardous Building. The Writer Robot enters first, explores without a pre-existing map, detects safety-relevant conditions, and leaves information behind in beacons. The information can later be recovered through the Outside Network and used to brief an Executor Robot.

2. Proposed system in one view

Subsystem	Hardware	Role / status
Writer Robot	Arduino UNO Q; RPLIDAR, Moduino Movement (LSM6DSOX IMU); Moduino Distance (VL53L4CD ToF); servo + beacon dispenser; thermal sensor; candidate eMLX90640 or AMG8833, not selected; wheeled drive base, wheel encoders, motor driver, battery and emergency stop to be finalized.	Autonomous exploration, localization, sensing, event detection, beacon decision and log. Core board/sensors confirmed by user; final mechanical/thermal choice open.
Beacon Network	ESP32-class MCU; SX1276/RFM95-class LoRa transceiver; non-volatile memory; LiPo battery; regulator; deployment switch; status LED; antenna; Reference architecture.	Persistent spatial memory, RF broadcast and beacon-to-beacon relay. Physical architecture proposed; RF band, exact module, protocol and range remain open.
Outside Network Area	Raspberry Pi gateway + GPS receiver; local network interface; MQTTServer layer; PCbrowser for command-post map; wireless or satellite backhaul.	RECEIVE, TRANSLATE, CARRY, BRIEF Architecture designed; final gatewaylink hardware open.
Command Post	PC/aptop + display/browser for the map/dashboard.	Visualize GPS-referenced events/path and select then. Proponed.
Executor Robot	Proposed reuses: UNO Q, RPLIDAR, Moduino Movement(IMU); floor-distance sensing; selected thermal sensor; beacon-compatible RF receiver; drive base + encoders + motor driver; battery; optional transmitter.	Enter later: listen to beacons, verify information, navigate and execute the mission. Architecture designed; hardware not finalized.

Core idea If the Writer fails, the robot may disappear - but the knowledge already stored in deployed beacons remains.

3. Design principles

- GPS-free inside: the Writer and Executor operate in a private coordinate frame referenced to the entrance.
- Persistent memory: beacons stay active after deployment and carry compact WHAT / WHERE / WHEN information.
- No direct inside-to-command-post link: all outside communication passes through the Outside Network Area.
- Resilience by design: Writer log recovery is used when the Writer returns; deployed beacons remain the fall-back when it does not.
- Honest prototyping: simulation results are separated from physical hardware validation and open engineering decisions.

4. Writer Robot - entering and exploring

The Writer is the first robot. It starts at the entrance, without a pre-existing map. The current implementation uses a hardware abstraction layer so the decision logic is independent from the physical hardware interface. Planned hardware: Arduino UNO Q; RPLIDAR, Modulino Movement (LSM6DSOX), Modulino Distance (VL53L4CD ToF), servo-based beacon dispenser, and a thermal sensor to be selected (MLX90640). A wheeled drive base, wheel encoders, motor driver, battery and emergency stop are also planned, with exact parts still to be finalized.

Stage	What the Writer does
1. Calibrate	Stops at the entrance for IMU bias calibration, then starts autonomous exploration.
2. Explore	Uses LIDAR-based wall following, obstacle checks and local position estimation.
3. Observe	Reads LIDAR, thermal and downward distance information to detect relevant events.
4. Decide	scores event candidates and decides whether a beacon is worth consuming.
5. Drop	Builds the beacon message, writes/configures the beacon, releases it and confirms the drop.
6. Continue / Recover	Continues exploration or executes a stuck-recovery maneuver when needed.
7. Return / Upload	Returns using a stored breadcrumb path when the mission ends, then uploads the detailed log near the entrance.

5. Navigation and localization

- Right-hand wall following: maintain approximately 0.40 m from the right wall; slow down while turning and react to front blockage or a missing floor.
- Movement estimate: wheel odometry + IMU heading. The simulation adds gyro bias, encoder scale error and measurement noise.
- Heading correction: a straight wall detected by LIDAR is used as a simplified stand-in for scan matching. The current simulation assumes an axis-aligned, Manhattan-style building.
- Return path: the Writer stores a breadcrumb trail and erases loops so the stored route remains a short known path back to the entrance.

6. Writer state machine

State	Purpose
CALIBRATE	Initial sensor calibration.
EXPLORE	Navigation + sensing + event evaluation.
DROP	Stop, configure, release and confirm beacon.
RECOVER	Back up and turn after a stall.
RETURN	Retrace the stored route to the entrance.
UPLOAD	Send the detailed log to the Outside Network.
DONE	Mission completed.
SAFE STOP	Safety state for tip-over or unrecoverable conditions.

Implementation boundary The Writer simulation is the most mature software component. The physical RPLIDAR, drive base, thermal sensor, beacon radio and dispenser remain to be integrated and validated.

7. Event detection and beacon decision

The challenge requires at least two event types. Our Phase 1 design uses two primary safety events plus a navigation landmark: HAZARD (heat/fire hotspot), HOLE (collapsed floor), and JUNCTION (branching point).

Type	Detection method	Why it matters	Beacon content
HAZARD	Thermal reading reaches the configured heat threshold; the Writer tracks the peak.	Preserves the location and severity of a possible fire/heat source.	Type, bearing, estimated distance, severity, time.
HOLE	Down-facing distance sensor detects a floor discontinuity above the configured threshold.	Prevents the Executor from blindly crossing a dangerous missing floor area.	Type, direction toward the hazard, time.
JUNCTION	LiDAR identifies multiple open directions and confirms the condition across consecutive scans.	Preserves a navigation landmark and available exits for the inherited map.	Type, direction, time, 4-bit exits mask.

8. Which beacon is dropped?

The Writer treats beacon stock as a limited resource. Each candidate event receives a score based on event importance, severity, novelty and remaining stock. When stock decreases, the acceptance threshold increases, so high-value events are protected while low-value relay/waypoint candidates can be skipped.

Example candidate	Current policy
HAZARD	High priority, severity increases the score.
HOLE	High fixed priority because it is a safety-critical obstacle.
JUNCTION	Medium priority; useful as a structural landmark.
WAYPOINT	Lower priority; used when beacon spacing would otherwise break the communication chain.

9. Beacon message and deployment sequence

- The current message proposal follows the challenge requirement that a beacon communicate what was found, where it is relevant, and when it was recorded.

Field	Meaning
id	Unique beacon identifier.
type	HAZARD / HOLE / JUNCTION / WAYPOINT.
dir_deg	Direction in the private frame.
dist_m	Event distance or a navigation distance, depending on event type.
t	Timestamp used for aging.
ver	Message version for later updates.
exits	4-bit open-direction mask for junctions.

- Deployment sequence: detect event → score → create message → configure beacon → servo release → confirm physical drop → beacon becomes active.
- Beacon architecture: an ESP32-class controller, non-volatile memory, independent battery/regulator, deployment switch, status LED, antenna and a reference SX1276/RFM95-class LoRa transceiver. This is the proposed physical hardware architecture; the exact RF technology, frequency, antenna and byte-level protocol remain team decisions.
- Proposed resilience upgrade: neighboring beacons may advertise previous/next beacon relationships and relay messages toward the Outside Network.

Open interface decision The physical beacon network and its exact packet protocol must be agreed with the 8. Beacon teammate before hardware implementation.

10. Beacon Network - persistent spatial memory

The beacon network is the part that allows knowledge to survive the Writer. A hybrid communication model is proposed: a beacon can communicate with the Outside Network when reachable, and neighboring beacons can relay information when a direct link is unavailable.

Mode	Flow	Purpose
Direct	Beacon → Outside Network	Fast delivery when a deployed beacon can reach the gateway.
Relay	Beacon 4 → Beacon 3 → Beacon 2 → Beacon 1 → Outside Network	Extends communication into disconnected areas.
Executor	Beacon → Executor	Provides local navigation and event information when the second robot enters.

11. Beacon chain proposal

At deployment time, a Writer cannot know the future location of the next beacon. We therefore propose that each newly deployed beacon identifies its predecessor, and that the previous beacon learns the successor through a short-range/local RF exchange. This creates a chain such as B1 → B2 → B3 without requiring future positions to be known in advance.

Stored navigation metadata	Purpose
previous_id / next_id	Identifies neighboring beacons in the chain.
direction / distance to neighbor	Supports local navigation between anchors.
message_id / hop_count / TTL	Prevents repeated forwarding and infinite relaying.
version	Allows a later update to supersede older information.

12. Outside Network Area

The Outside Network is the bridge between the disconnected interior and the outside world. The proposed hardware is a Raspberry Pi gateway with a GPS receiver at the entrance, a local network interface and MQTT server services, plus a PC/browser at the command post for the map. The final wireless or satellite backhaul remains to be selected.

Role	Input	Output
RECEIVE	Writer log I3 and/or recovered beacon information	Validated stored data
TRANSLATE	Private coordinates + entrance GPS + building bearing	GPS-referenced events/path + uncertainty
CARRY	GPS-referenced data	Command-post update; store-and-forward if link is down
BRIEF	Mission + translated data + beacon chain	Executor briefing I5

13. Frame translation

The Writer uses a private frame whose origin is the entrance. The Outside Network combines the entrance GPS fix and the building entry bearing to transform each point into East/North displacement and then latitude/longitude. The map should preserve an uncertainty estimate because dead-reckoning and bearing errors grow with distance.

Communication rule The command post never connects directly to the robots. Inside-to-outside communication passes through the Outside Network Area.

14. Executor Robot - entering with inherited knowledge

The Executor is the second robot. Planned/proposed hardware reuses the Writer architecture where practical: UNO Q, RPLIDAR, Modulino Movement/IMU, floor-distance sensing, the selected thermal sensor, a beacon-compatible RF receiver, drive base with encoders and motor driver, battery, and: an optional transmitter. Exact Executor hardware is not yet finalized.

State	behaviour
BRIEFED	Receives mission, beacon chain, known hazards, clock information and trust rules.
ENTER	Starts from the same entrance convention and establishes its private frame.
LISTEN	Moves toward the expected next beacon while monitoring its own sensors.
TRUST-CHECK	Evaluates beacon age, version and agreement with onboard sensors.
FOLLOW	Uses the beacon chain for local navigation.
ACT	Performs the mission at the target, e.g. reach a heat hazard for an extinguishing task (simulated in Phase 1).
EXIT	Leaves the building by retracing its route; exact exit logic remains to be finalized.
SAFE STOP	Stops for unrecoverable or unsafe conditions.

15. Trust by age

Beacon status	Executor interpretation
Fresh	Trust more strongly and follow at normal speed.
Aging	Slow down and verify with own sensors.
State	Treat as guidance only; rely primarily on onboard sensing.
Contradicted / suspect	Flag and discard the conflicting information.

16. End-to-end mission flow example

- 1. Writer enters a fire-damaged building with no GPS and explores autonomously.
- 2. LIDAR detects a junction; the Writer records the available exits and may place a JUNCTION beacon.
- 3. The thermal sensor identifies a heat hotspot; the Writer records HAZARD information and places a beacon when the policy accepts it.
- 4. A missing-floor condition is detected; the Writer stops/turns away and records a HOLE event.
- 5. Deploys beacons remain active and can relay information toward the Outside Network.
- 6. If the Writer returns, its detailed log is received by the Outside Network and translated to GPS; if it fails, beacon memory remains the fallback.
- 7. The command post selects the mission and the Outside Network briefs the Executor.
- The Executor enters, listens for beacons, checks their age and sensor consistency, follows the inherited route, reaches the target and exits.

Resilience mechanism Two complementary memory paths: detailed Writer log when the Writer returns, and persistent beacon memory when the Writer does not.

17. Phase 1 simulation and evidence

The current Writer simulation is implemented in Python and models a hazardous building with LIDAR, thermal sensing, a floor sensor, odometry/gyro imperfections, battery drain, beacon decisions and logging. The simulation is hardware-independent through the WriterHAL interface. The simulated interfaces correspond to the planned LIDAR, thermal, floor-distance, IMU/odometry, battery and beacon-dispenser functions; the physical components are not yet integrated.

Writer run - status: returned | 5 beacons | path 55 m

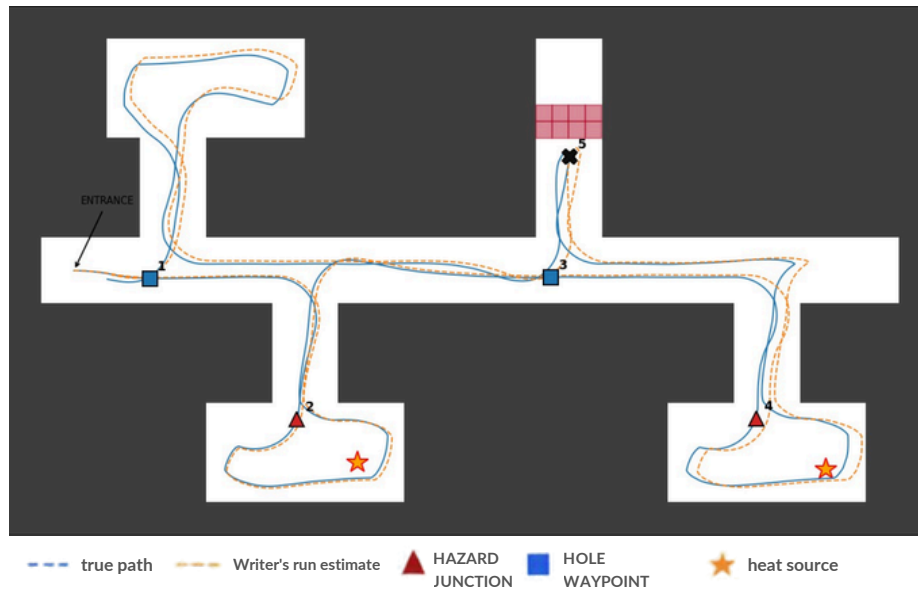


Figure 1. Writer simulation example: trajectory, estimated path and deployed beacons.

Current evidence	Observed result	Interpretation
Normal run (seed 0)	Returned: 55.1 m path; 5 beacons; log uploaded; final dir of 0.09 m.	Demonstrates end-to-end Writer simulation flow.
14-seed robustness script	14 /14 runs returned with HAZARD + HOLE + JUNCTION detected.	Current clean-run criterion passes, it does not independently verify full building coverage.
Failure scenarios	Command-line hooks exist for Writer death, drop failure, low battery and debris.	Execution and screenshots still need to be produced for final Phase 1 evidence.

18. Failure cases and implementation plan

Failure	Expected response	Phase
Writer failure / loss	No detailed beacons already deployed remain usable.	Phase 1 demo + Phase 2 validation
Beacon/drop failure	Retry: if unsuccessful, log failure and continue.	Phase 1 demo + hardware test
Low battery	Stop exploration and return using stored path.	Phase 1 demo + hardware test
Stuck / obstacle	Recover by backing/turning: repeated stalls trigger return.	Phase 1 demo + hardware test
Outside link loss	Queue data and forward when the link returns.	Phase 1 simulation + physical later

Now - Phase 1	After submission - Final Phase
Freeze 11-15 interfaces and event set	Finalize RF technology and beacon electronics
Integrate Writer log → Outside Network → Executor simulation	Build physical Writer / Beacon / Outside Network / Executor
Generate clean demo, failure evidence and diagrams	Validate navigation, RF range, GPS translation and trust/aging
Finalize GitHub + <6 page report	Complete prototype, user manual and final pitch